

Customer Shopping Behaviour Analysis

1. Project Overview

This project focuses on analyzing customer shopping behavior using a transactional dataset comprising approximately 3,900 purchase records across multiple product categories. The primary objective is to derive actionable insights into customer spending habits, segmentation patterns, product preferences, and subscription trends, ultimately supporting data-driven business decision-making.

2. Dataset Description

- **Rows:** 3,900
- **Columns:** 18

Key Attributes Include:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Transaction Details:** Product Name, Category, Purchase Amount, Season, Size, Colour
- **Behavioural Metrics:** Discount Usage, Purchase Frequency, Previous Purchases, Review Ratings, Shipping Preferences

Data Quality Note:

- Identified 37 missing values in the *Review Rating* column, which were addressed during pre-processing.

3. Data Pre-processing & Exploratory Data Analysis (Python)

Data cleaning and exploratory analysis were conducted using Python to ensure data integrity and usability.

- **Data Loading:** Loaded dataset using **Pandas** for structured analysis.

	customer_id	age	gender	item_purchased	category	purchased_amount	location	size	color	season	review_rating	subscription_status	shipping_type
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	2.5	Yes	Next Day Air
4	5	45	Male	Blouse	Clothing	48	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping

- **Initial Assessment:** Evaluated dataset structure using `.info()` and statistical summaries via `.describe()`.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

```
df.isnull().sum()
[9]
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
```

- **Handling Missing Values:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake_case** for better readability and documentation.

- **Feature Engineering:**

- Created an **age_group** variable by categorizing customers into age brackets.

	age_group	age
0	Middle-aged	55
1	Young Adults	19
2	Middle-aged	50
3	Young Adults	21
4	Middle-aged	45
5	Middle-aged	46
6	Senior	63
7	Young Adults	27
8	Young Adults	26
9	Middle-aged	57

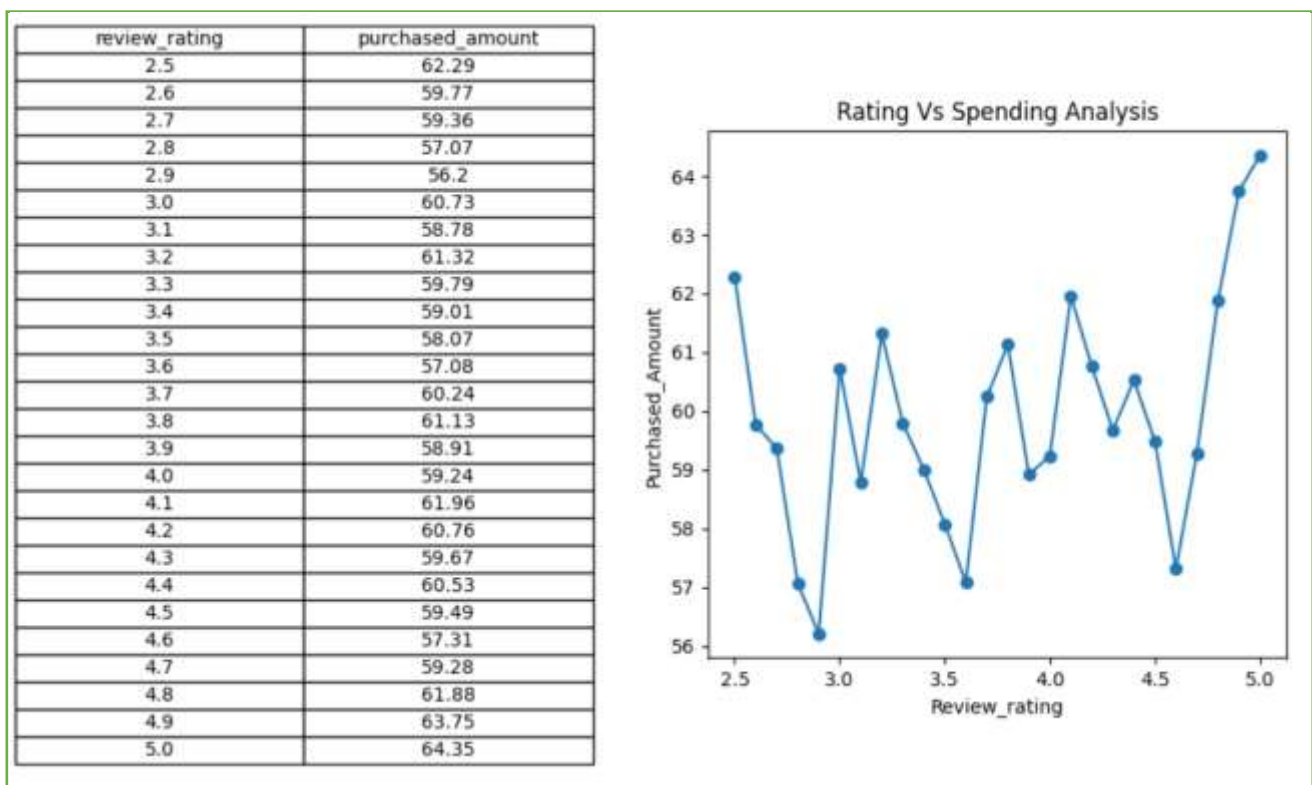
- Derived **purchase_frequency_days** to quantify customer buying intervals.

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14
1	Fortnightly	14
2	Weekly	7
3	Weekly	7
4	Annually	365
5	Weekly	7
6	Quarterly	90
7	Weekly	7
8	Annually	365
9	Quarterly	90

Payment Method Analysis: Identified which payment method is more utilised for the purchase

	payment_method	purchased_amount
2	Credit Card	40310
4	PayPal	40109
1	Cash	40002
3	Debit Card	38742
5	Venmo	37374
0	Bank Transfer	36544

Rating Vs Spending Analysis :



- **Data Validation:**

Identified redundancy between **discount_applied** and **promo_code_used**, leading to the removal of the less informative feature.

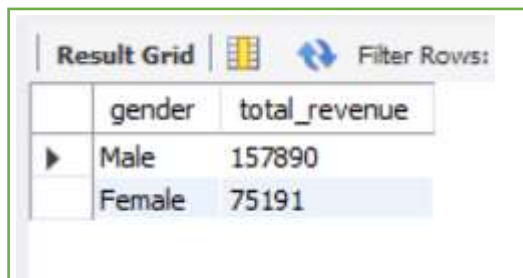
- **Database Integration:**

The cleaned dataset was exported and stored in a MySQL database to facilitate advanced querying and analysis.

4. Business Analysis Using MySQL

Advanced analytical queries were performed in MySQL to extract key business insights:

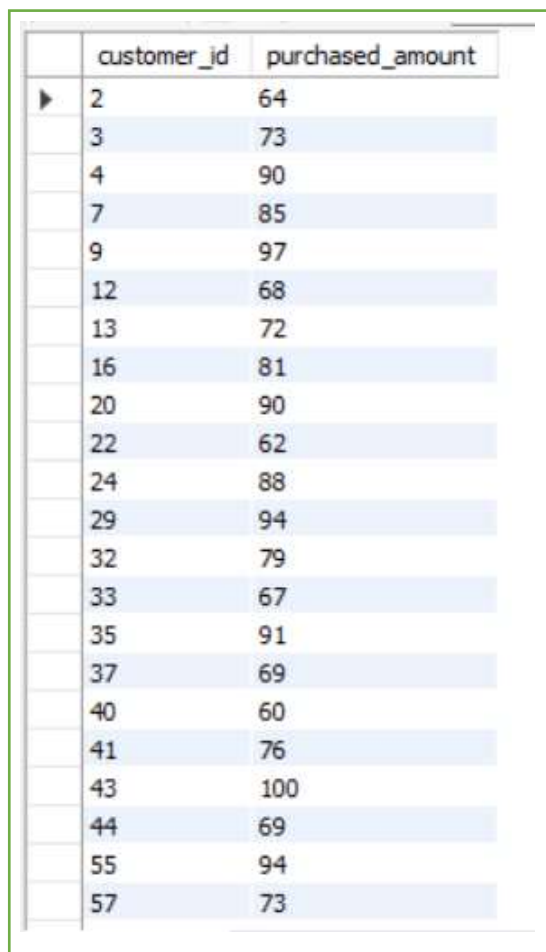
1. **Revenue Distribution by Gender:** Compared total revenue contribution across male and female customers.



A screenshot of a MySQL Result Grid interface. At the top, there are tabs for 'Result Grid', a grid icon, and a 'Filter Rows:' button. Below the tabs is a table with two columns: 'gender' and 'total_revenue'. The table contains two rows: 'Male' with a value of 157890 and 'Female' with a value of 75191. The 'Female' row is highlighted in light blue.

	gender	total_revenue
▶	Male	157890
	Female	75191

2. **High-Value Discount Users:** Identified customers who utilized discounts yet maintained above-average spending levels.



A screenshot of a MySQL Result Grid interface showing a list of customers. The table has two columns: 'customer_id' and 'purchased_amount'. The first row is highlighted in light blue. The table contains 20 rows of data.

	customer_id	purchased_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69
	40	60
	41	76
	43	100
	44	69
	55	94
	57	73

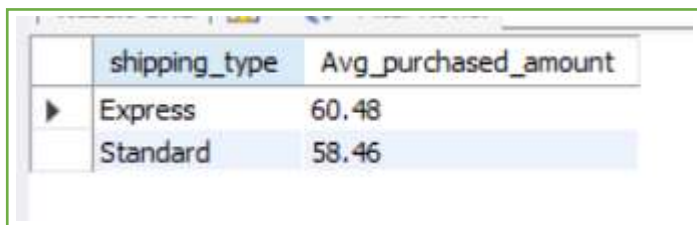
3. **Top-Rated Products:** Ranked products based on average customer review ratings.



A screenshot of a data table with two columns: 'item_purchased' and 'Avg_Product_Rating'. The table lists five items: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.8), and Handbag (3.78). The rows are alternatingly highlighted in white and light blue.

item_purchased	Avg_Product_Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

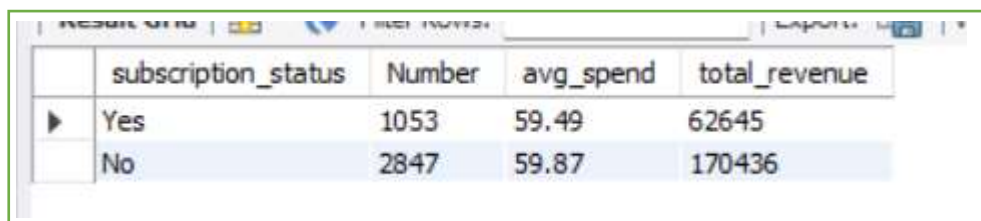
4. **Shipping Preference Analysis:** Evaluated differences in purchase value between *Standard* and *Express* shipping options.



A screenshot of a data table with two columns: 'shipping_type' and 'Avg_purchased_amount'. The table lists two shipping types: Express (60.48) and Standard (58.46). The rows are alternatingly highlighted in white and light blue.

shipping_type	Avg_purchased_amount
Express	60.48
Standard	58.46

5. **Subscription Impact Analysis:** Assessed spending patterns and revenue contribution of subscribers versus non-subscribers.



A screenshot of a data table with four columns: 'subscription_status', 'Number', 'avg_spend', and 'total_revenue'. The table lists two subscription statuses: Yes (1053, 59.49, 62645) and No (2847, 59.87, 170436). The rows are alternatingly highlighted in white and light blue.

subscription_status	Number	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

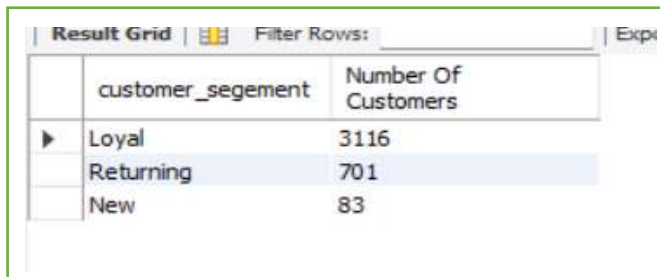
6. **Discount Sensitivity Analysis:** Determined products most frequently purchased under discount conditions.



A screenshot of a data table with two columns: 'item_purchased' and 'discount_rate'. The table lists five items: Hat (50.00), Sneakers (49.66), Coat (49.07), Sweater (48.17), and Pants (47.37). The rows are alternatingly highlighted in white and light blue.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation:** Classified users into *New*, *Returning*, and *Loyal* categories based on purchase frequency.



The screenshot shows a 'Result Grid' with a 'Filter Rows' section and an 'Export' button. The table has two columns: 'customer_segement' and 'Number Of Customers'. The data is as follows:

	customer_segement	Number Of Customers
▶	Loyal	3116
	Returning	701
	New	83

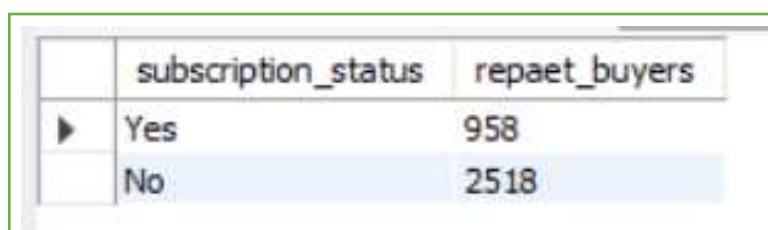
8. **Category-Level Product Performance:** Identified top-performing products within each category.



The screenshot shows a table with five columns: 'item_rank', 'category', 'item_purchased', and 'total_orders'. The data is as follows:

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

9. **Repeat Purchase Behaviour:** Analyzed correlation between high purchase frequency (>5 purchases) and subscription likelihood.



The screenshot shows a table with two columns: 'subscription_status' and 'repaet_buyers'. The data is as follows:

	subscription_status	repaet_buyers
▶	Yes	958
	No	2518

10. Revenue by Age Segment: Measured contribution of different age groups to overall revenue.

	age_group	total_revenue
▶	Young Adults	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Data Visualization (Power BI Dashboard)

An interactive dashboard was developed using Power BI to present insights in a visually intuitive format.

The dashboard includes:

- Revenue trends and KPIs
- Customer segmentation visuals
- Product performance analysis
- Behavioural insights (discount usage, shipping trends)



6. Strategic Recommendations

Based on the analysis, the following business strategies are recommended:

- **Enhance Subscription Programs:**
Introduce exclusive incentives to increase subscriber conversion rates.
- **Strengthen Customer Retention:**
Implement loyalty programs targeting repeat customers to transition them into high-value segments.
- **Optimize Discount Strategy:**
Balance promotional offers to maximize revenue without compromising profit margins.
- **Improve Product Marketing:**
Highlight top-rated and best-selling products in promotional campaigns.
- **Targeted Customer Engagement:**
Focus marketing efforts on high-revenue demographics and customers preferring express delivery.